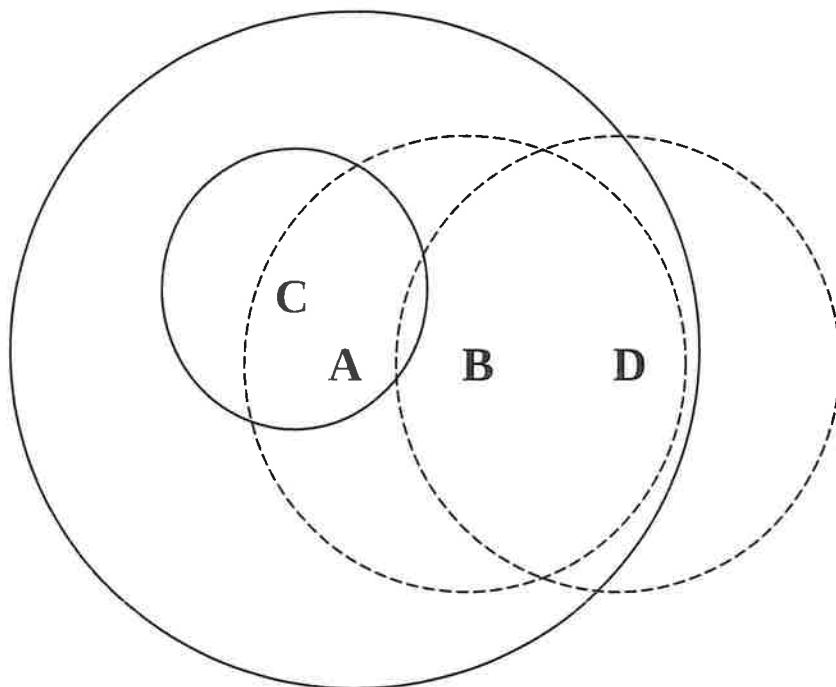


CS348 Computer Networks Quiz-1, Thu 27 Aug 2026, 09.30am-10.10am, Max. Marks: 10Name: SAMPLE ANSWERS Roll number: _____ House: _____**General instructions**

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only. You do NOT need a calculator.
- All answers have to be (briefly) explained. State any necessary assumptions.
- The final answer of each question has to be entered in the given box.

Consider the following wireless network for all questions below. The transmit range of each node is shown with a circle whose center is marked with the node name in the diagram. Assume that this is a LAN within an office building.



A's transmission can be heard by B, C, D. B's transmission can be heard by A, C, D. C's transmission can be heard only by A. D's transmission can be heard only by B.

Q1 [2 marks]: In A-to-B communication, suppose a stop-and-wait protocol is used at the link layer. Assume a transmission speed of 5Mbps, a data packet size of 1000 bytes (payload from network layer), and an ACK size of 10 bytes. The link layer header is 10 bytes, and trailer is another 10 bytes. What is the efficiency of the stop-and-wait protocol? Give your answer to the nearest integer percentage. Show the relevant steps. Make (and briefly explain) engineering approximations.

A1 steps and explanation:

Propagation delay is negligible e.g. $\frac{300m}{3 \times 10^8 m/s} \approx 1\mu s$

$$T_x \text{ delay} = \frac{1000}{5M} = 1.6 \text{ ms}$$

So efficiency loss only due to header, trailer, ACK

$$Eff = \frac{1000}{1000+10+10+10} = \frac{100}{103} \approx \frac{97}{100} = 97\%$$

A1 final answer (nearest integer percentage, give only the number here):

97

Q2 [2 marks]: Suppose the wireless network used WiFi MAC along with a retransmission scheme with $SWS=RWS=4$. How many *bits* are needed to represent the sequence number in the ACK packet? *Hint*: Recall the gaps in the MAC protocol; the answer needs you to think from first principles.

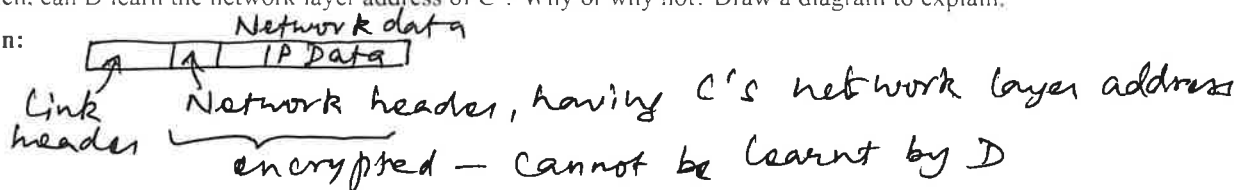
A2 explanation: ACK comes immediately after SIFS after data
So no confusion about which packet is being ACKed
So no need for any seq. num. field in ACK

0

A2 final answer (integer, give only the number here):

Q3 [2 marks]: Continue to assume that the network uses WiFi. When A is sending a network layer packet to C, it uses link-layer encryption using a key known only to A and C. Assuming the cryptographic algorithm is strong and cannot be broken, can D learn the network layer address of C? Why or why not? Draw a diagram to explain.

A3 explanation:



N

A3 final answer (write Y or N):

Q4 [3 marks]: Continue to assume that the network uses WiFi MAC. Now, D sends broadcast frames as fast as allowed by the MAC, at a transmit rate of 5Mbps. And C sends broadcast frames as fast as allowed by the MAC, at a transmit rate of 1Mbps. There is no other traffic in the network. What is the sum total of received throughput at all the 4 nodes? Answer in Mbps.

A4 brief steps and explanation: Broadcast $\Rightarrow$ no ACK

C, D do not see each other, so transmit continuously
Ignoring MAC backoff overheads, sum total is $D \rightarrow B$ 5Mbps
+ $C \rightarrow A$ 1Mbps = 6Mbps
No interference either, at receivers for $D \rightarrow B$ and $C \rightarrow A$

6

A4 final answer (number, in Mbps):

Q5 [1 mark]: In this network, is a centralized polling based MAC possible? Why or why not?

A5 explanation: No node can transmit ~~AND~~ AND receive from all other nodes. So not possible.

N

A5 final answer (write Y or N):